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Editors
Science

Dear Editors,

Please consider my manuscript, “On the Basis of Mathematics,” for publication in *Science*.

The manuscript proposes “writability”—the capacity of an idea or statement to be instantiated in communicable symbols—as a fundamental bound on mathematical language. Beginning with self-reference and the liar paradox, it develops a minimal representational framework from which reference, truth values, logical connectives, and a language of discourse can be constructed. It then examines a provocative consequence: if every mathematically usable expression must be writable, the conventional separation between object-language and meta-language may require reconsideration. From this perspective, the manuscript argues that mathematical completeness and incompleteness can be understood as simultaneously arising within a self-referential language.

This work addresses foundational questions extending across mathematical logic, the philosophy of mathematics, theories of computation, and the scientific study of language and representation. Its principal contribution is to identify writability as an explicit constraint on formal inquiry and to use that constraint to offer a unified interpretation of self-reference, Gödelian completeness and incompleteness, and Hilbert’s second problem. Because the argument concerns the representational foundations shared by mathematics and the sciences, I believe it may interest *Science*’s broad interdisciplinary readership.

I am the sole author and corresponding author:

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This manuscript has not been published previously and is not under consideration by another journal. No closely related manuscript is under consideration or in press elsewhere.

I have no competing interests to declare.

For transparency, ChatGPT was used in developing certain examples and the idea of employing photographs to illustrate self-reference, as acknowledged in the manuscript. ChatGPT was also used as a writing aid in preparing this cover letter. I take full responsibility for the manuscript’s

arguments, claims, presentation, and submitted text.

Thank you for considering this work. I would welcome the opportunity to have it evaluated by experts in mathematical logic, foundations of mathematics, and the philosophy of formal systems.

Sincerely,

Christopher Robinson Tompkins III
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